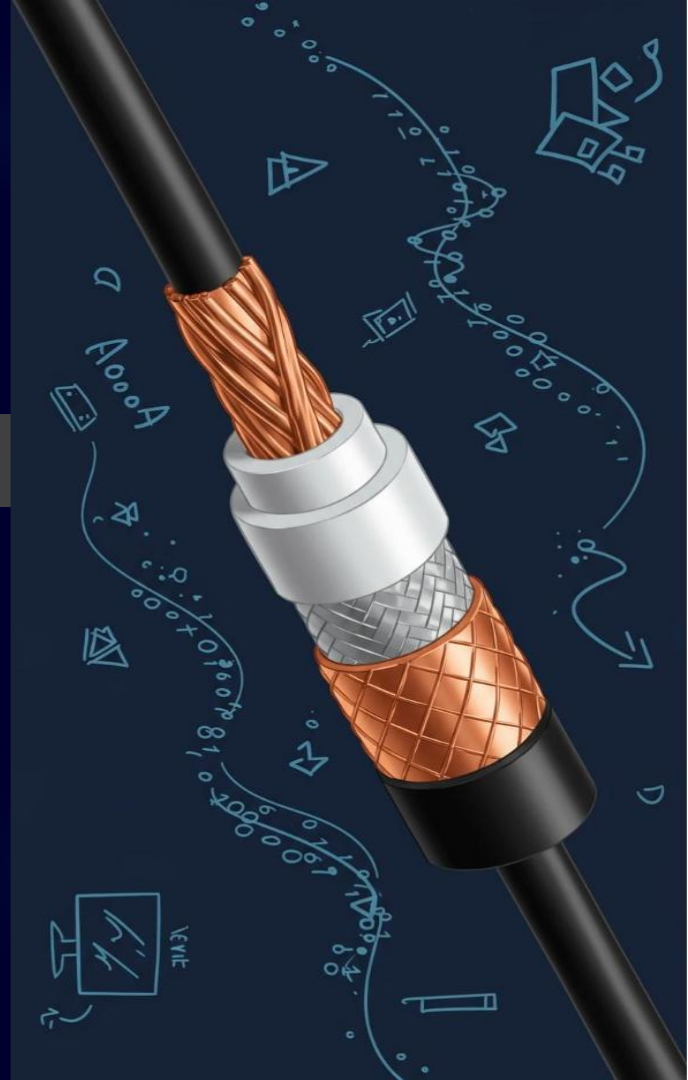


Coaxial Cable

As a Transmission Medium



Group 2

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Ayesha Hasib

Lamia



Presentation Roadmap

Transmission Medium
Overview

Advantages

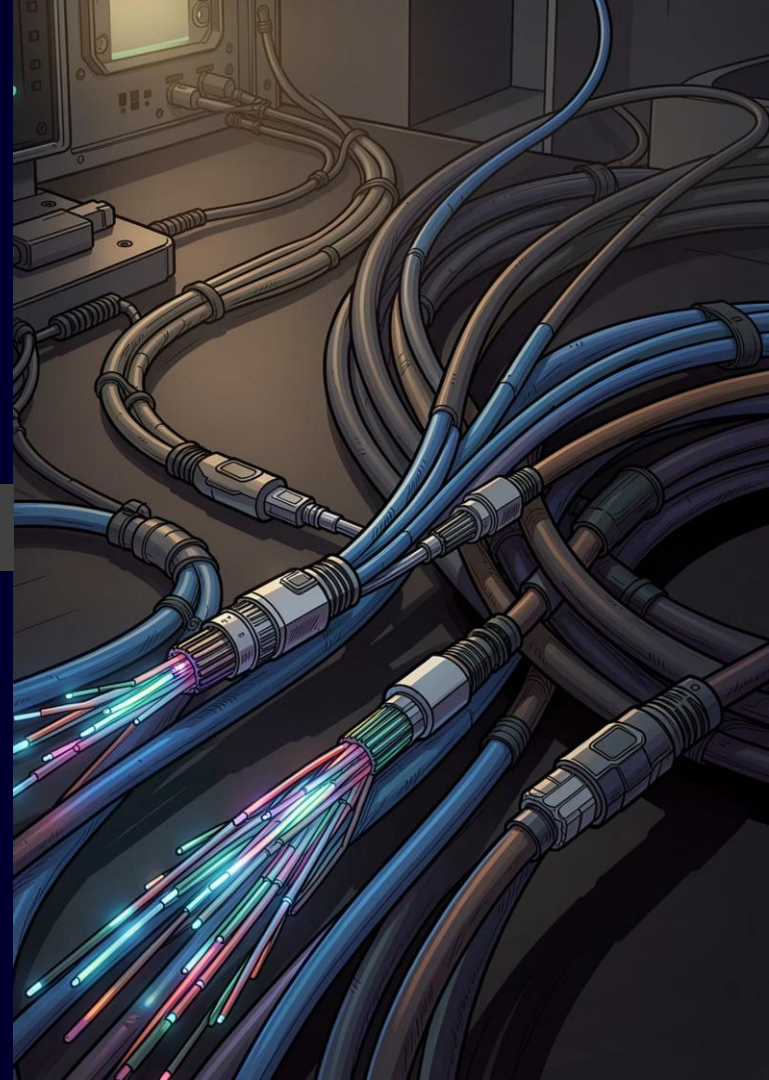
Real-Life Examples
/ Applications

Transmission Method
/ Technique

Disadvantages



Transmission Medium Overview



Transmission Media

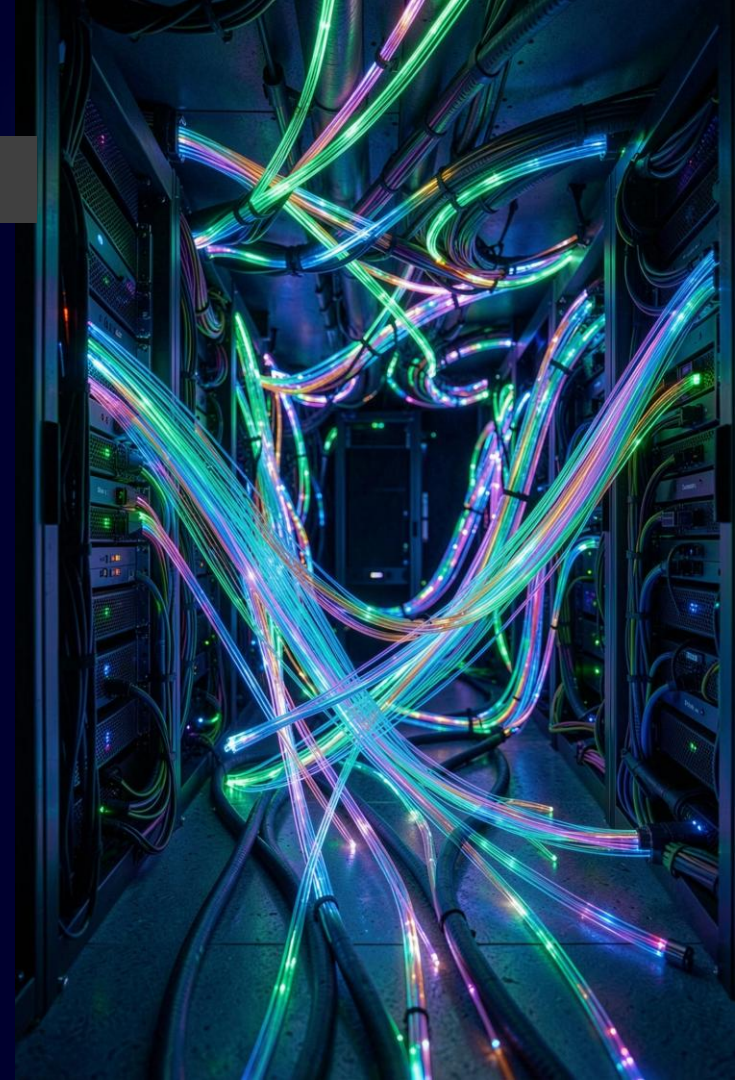
The physical or logical pathway that carries data signals between devices

What Is It?

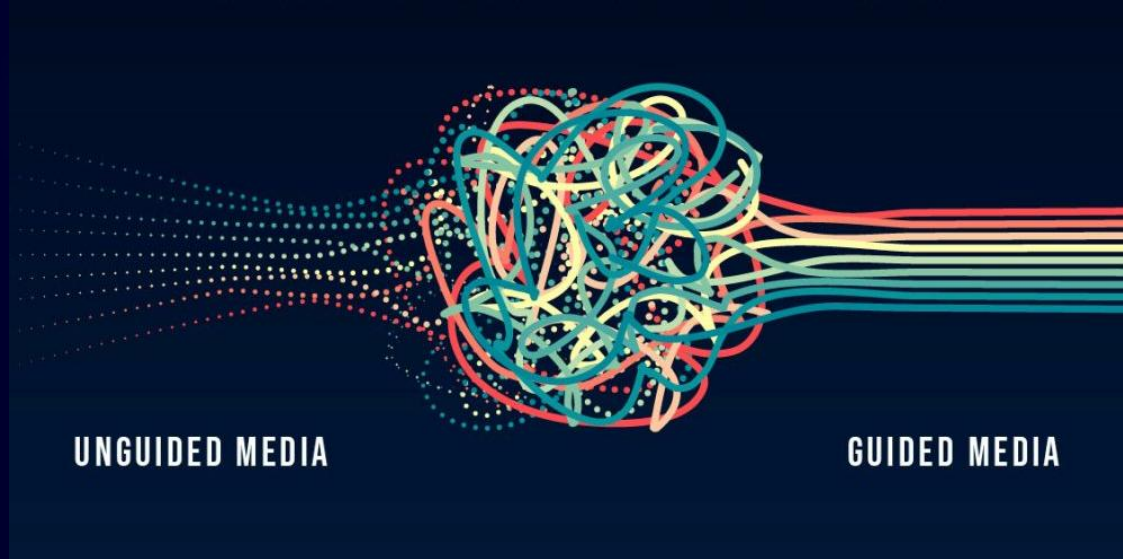
- A medium
- Physical or wireless
- Connects devices and enables data to flow from source to destination.

Why Does It Matter?

It directly determines the **speed, reliability, and security** of every digital communication.



Types of Transmission Media



GUIDED MEDIA

Physical pathways that direct signals along a fixed route

- **Coaxial Cable**
- **Twisted Pair Cable**
- **Fibre-Optic Cable**



UNGUIDED MEDIA

Wireless signals through air or vacuum

- **Radio Waves**
- **Microwaves**
- **Satellite**



Coaxial Cable

Coaxial Cable

Guided Transmission Medium

High-Frequency Signal Carrier

Coaxial Design for Shielding

Versatile and Durable



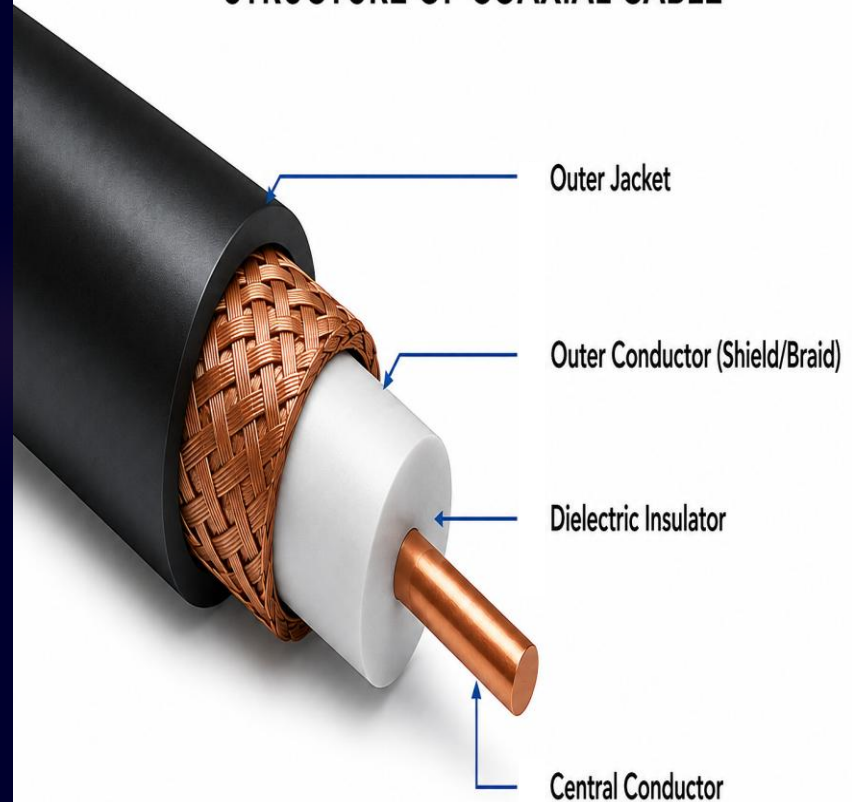
Structure of Coaxial Cable



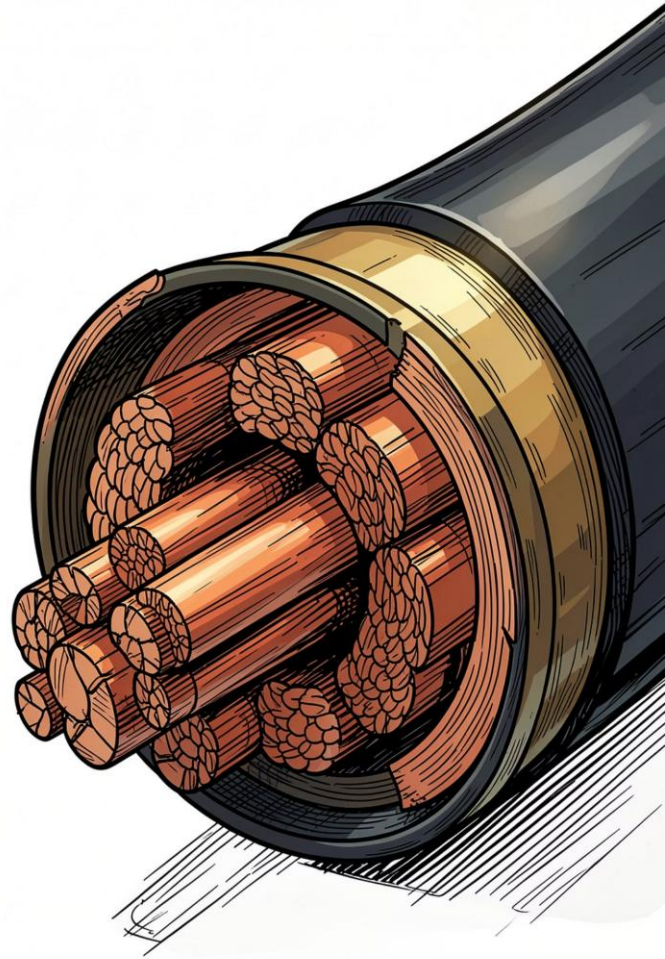
Structure of Coaxial Cable

- Inner Conductor
Copper Core – carries data signals
- Dielectric Insulator
prevents signal loss
- Metallic Shield
protects from electromagnetic interference
- Outer Jacket
protects the cable physically

STRUCTURE OF COAXIAL CABLE



Transmission Method / Technique



Data Origin & Line Coding



Digital Data is Created

A computer or device generates digital data in binary form — a continuous stream of **0s and 1s** representing all information: video, audio, internet traffic.

Line Coding Converts Bits to Voltage

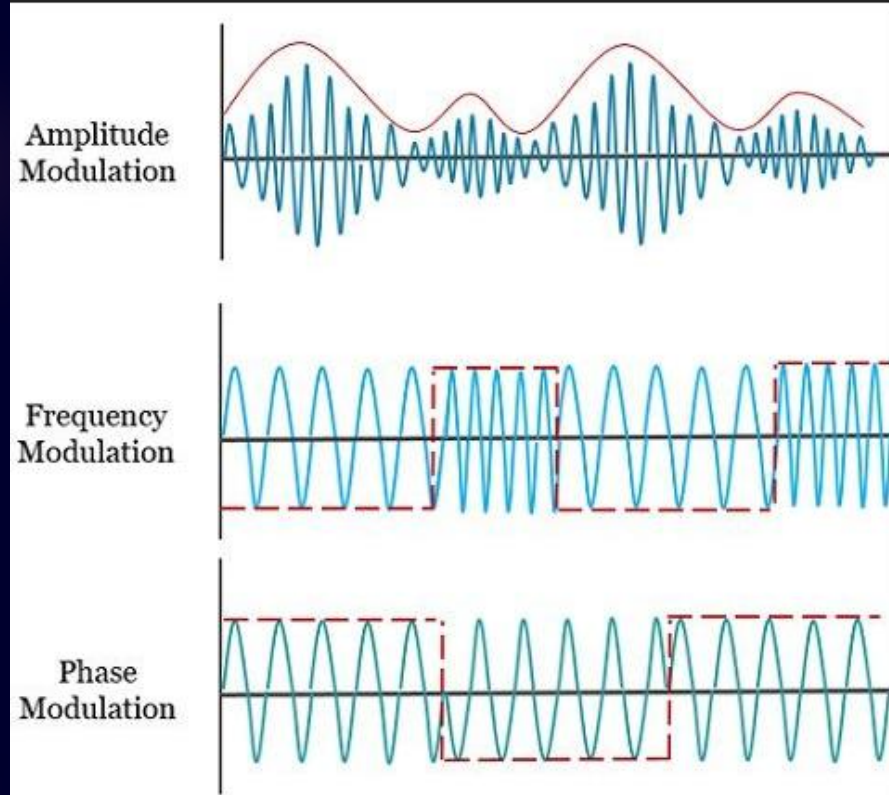
Binary bits are mapped to electrical signals:

- 1 → High voltage
- 0 → Low voltage

This electrical representation is what physically enters the cable.

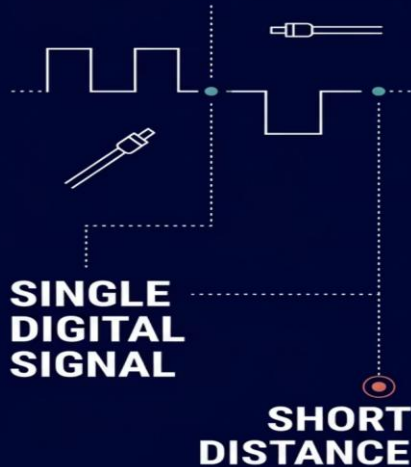
Modulation

Modulation changes electrical signals to travel efficiently over a communication channel.

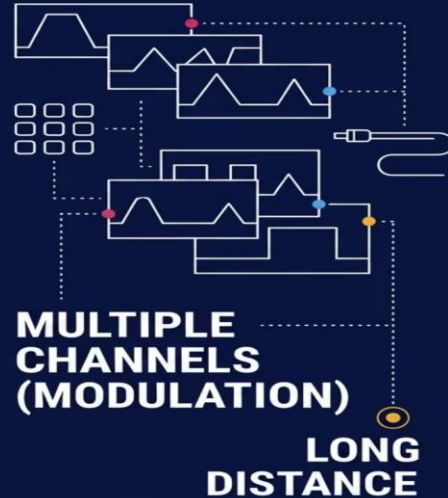


Transmission Technique

BASEBAND



BROADBAND



- Baseband Transmission
 - Entire cable used for one signal
 - Example: Ethernet networks
- Broadband Transmission
 - Multiple channels using modulation
 - Example: Cable TV and internet

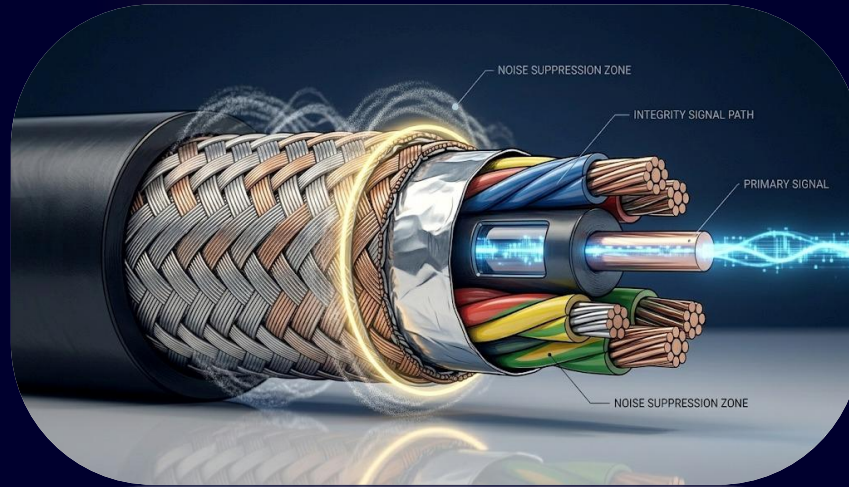
Signals Travel Through the Coaxial Cable



Coaxial Cable: Advantages



High Signal Quality



Noise Reduction

The metallic shield absorbs external interference before it reaches the signal conductor.

Stable Communication

Consistent signal integrity makes coax reliable for critical data transmission.

Better Than Twisted Pair

Outperforms unshielded twisted pair (UTP) cables in noisy environments.

High Bandwidth Capacity



Internet Services

Cable internet providers rely on coax to deliver broadband to homes and businesses with reliable throughput.



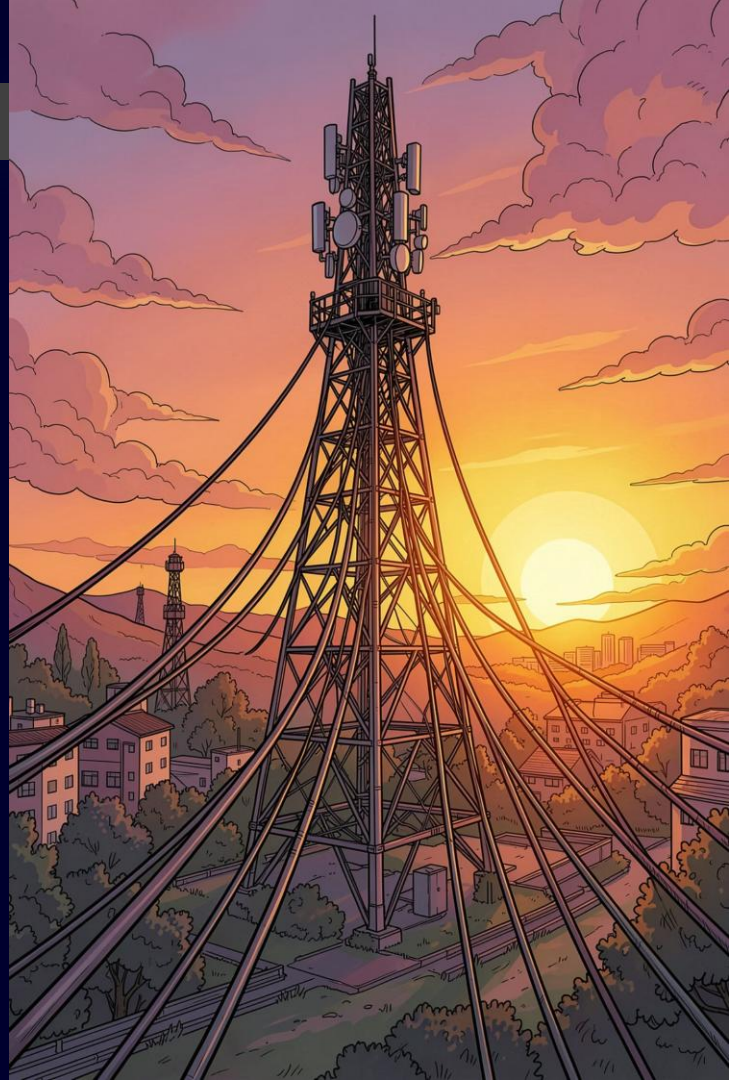
TV Broadcasting

Cable television systems use coaxial infrastructure to carry hundreds of channels simultaneously.



Video Transmission

Security systems and broadcast studios use coax for high-definition video feeds over short to medium distances.



Durable and Reliable

- **Long lifespan:** Coax installations can remain functional for decades with minimal maintenance.
- **Long-distance transmission:** Capable of carrying signals over hundreds of meters before requiring amplification or repeaters.
- **Environmental resilience:** Resistant to bending, temperature variation, and external physical impact.

Coaxial Cable: Disadvantages



Higher Cost

Material Cost

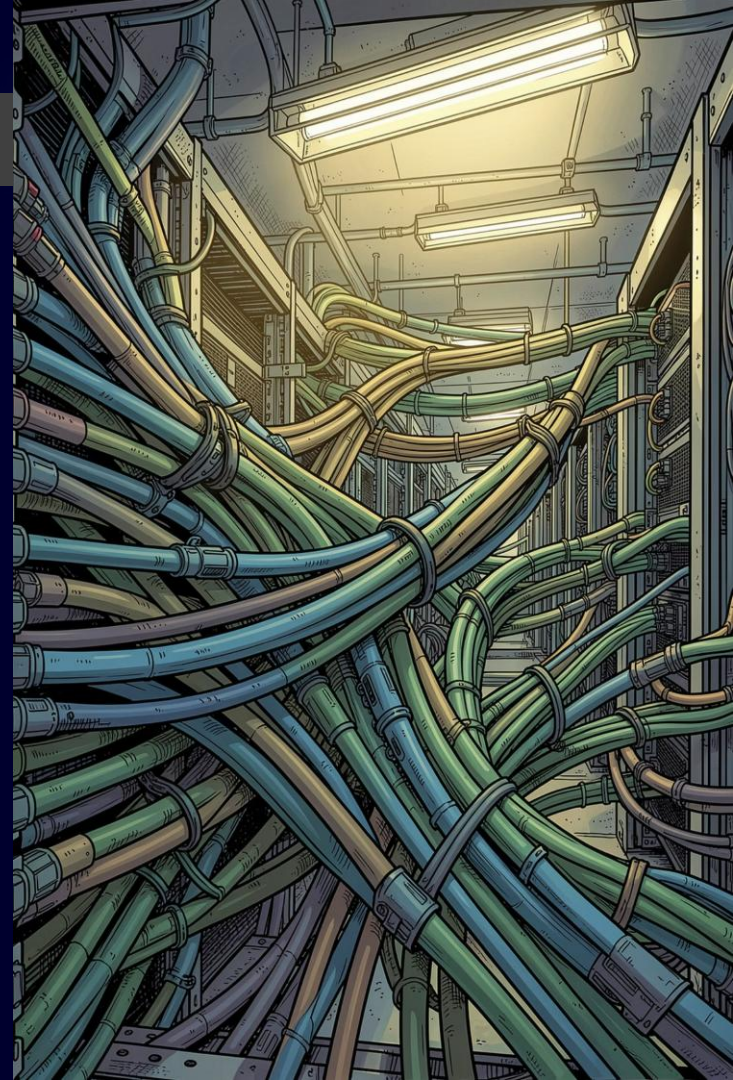
The multi-layer construction — copper core, insulation, shielding braid, and outer jacket — uses more raw material than simple twisted pair wire.

Installation Cost

Coaxial connectors (e.g., BNC, F-type) require more precise termination, increasing skilled labor time.

Maintenance Cost

Fault diagnosis and cable replacement in large coax networks can be time-consuming and costly.



Bulky and Less Flexible

- Thick and heavy cable
- Difficult to install in small spaces
- Less flexible than modern fiber cables



Limited Compared to Fiber Optics

- Lower speed than fiber optic cables
- Signal weakens over very long distances
- Not ideal for future ultra-high-speed networks



Real-Life Uses of Coaxial Cable





Cable Television

Signal Delivery

Carries both video and audio signals to your TV simultaneously

Wide Bandwidth

Supports hundreds of channels with minimal interference

Internet & Security Applications



Broadband Internet

Cable ISPs use coaxial lines to deliver high-speed internet to homes and offices — the same wire that carries TV also carries data.



CCTV & Security Systems

Surveillance cameras rely on coaxial cable to transmit video feeds to monitors and recorders — stable, long-distance, and interference-resistant.

Coaxial used for Cable Television System

- **TV Signals are Generated**
- **Signals are Modulated**
- **Signals Travel Through the Coaxial Cable**
- **Distribution to Homes**
- **TV Receives and Decodes the Signals**





THANK YOU

An illustration featuring five stylized people of various ages and genders holding up a large white rectangular sign with the words 'THANK YOU' in bold, dark blue capital letters. The sign is positioned in the upper center of the frame. Above the sign, there are decorative elements including white streamers and several white triangular confetti pieces. The background is a solid light blue color. The people are standing on a light blue ground plane. From left to right: a woman with long brown hair wearing a white long-sleeved shirt and dark blue pants; a man with short brown hair wearing a dark blue long-sleeved shirt and white pants; a woman with long brown hair wearing a teal long-sleeved shirt and a white skirt; a man with a beard and short brown hair wearing a dark blue jacket over a white shirt and black pants; and a man with short brown hair wearing a light green jacket over a white shirt and dark blue pants. All five individuals have their arms raised, holding the bottom edge of the sign.

ANY

QUESTION

